

TrES-3b

R-1808

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_150529 · created 2026-08-03T03:11:21+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2457171.8525 +/- 0.0011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.194 +/- 0.012
Transit depth (Rp/Rs)^2	3.76 +/- 0.48 [%]
Orbital Inclination (inc)	82.56 +/- 0.91
Airmass coefficient 1 (a1)	1.103 +/- 0.01
Airmass coefficient 2 (a2)	-0.094 +/- 0.062
Scatter in the residuals of the lightcurve fit is	0.95 %
Best Comparison Star	#2 - [501, 429]
Optimal Aperture	3.97
Optimal Annulus	6.67
Transit Duration (day)	0.0632 +/- 0.0069

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.194 ± 0.012 vs 0.16309 (deviation 2.58σ)
Duration fit vs published	0.0632 d vs 0.0567 d (deviation 0.95σ)
Mid-time O–C	-1.4 min
Depth SNR	16.2
Residual scatter	0.95 %

Light curve

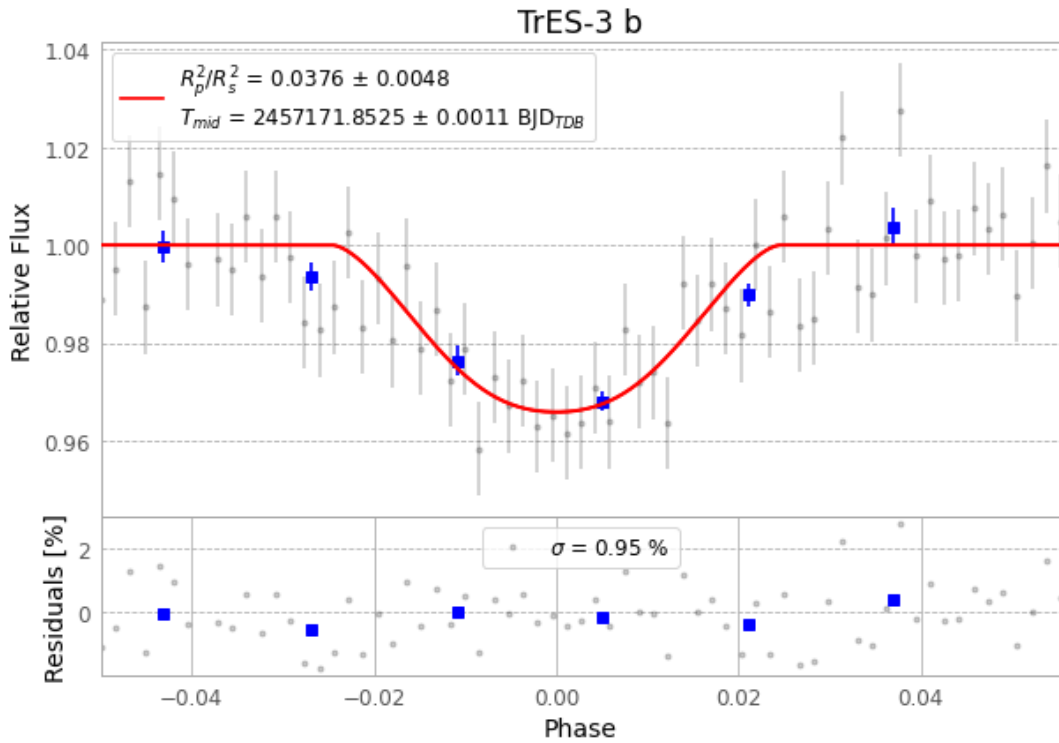


Figure 1 — FinalLightCurve_TrES-3 b_2015-05-28.png (SHA-256 9cb62289c301f0b9...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [259, 218], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 bb2b49699c5f48a2...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

67 FITS frames (44 MB), TRES-3150529064812.FITS → TRES-3150529100612.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-150529.log (be6c489422a8...), FinalLightCurve_TrES-3 b_2015-05-28.png (9cb62289c301...), FinalLightCurve_TrES-3 b_2015-05-28.csv (a4e87e4f701a...)

Compute resources

Wall time 230.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-03 03:11Z.